



SYMBIOSIS
SCHOOL OF ECONOMICS

**A Bibliometric Review of Global Research Trends in
Behavioral Economics and Financial Decision-Making using
Biblioshiny**

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	Conceptual Structure, Social structure, Formatting	Analysis of documents, Conclusion	Introduction, Sources, References, Formatting	Abstract, Authors, Clustering by coupling	Overview, Research Methodology

Subject: Research Methodology and Software
Packages 1

Submitted to: Dr. Sneha Kumari

Abstract

Background: Behavioural economics is an important field that explains how human psychology influences financial choices and often challenging the traditional view of people as rational decision makers. Biases such as overconfidence, loss aversion and present bias shape decisions about the savings, borrowing and investing. It also affects financial well being of the people. This has not only made behavioural economics highly relevant for individuals, businesses and policy

Purpose: The purpose of this study is to examine the global trends in behavioural economics and financial decision making through a bibliometric review. It aims to identify influential authors, countries, institutions and themes while highlighting how the research in this subject has developed over time.

Methodology: Data were collected from the Web of Science database for the period of 2015- 2025 using Boolean search terms such as behavioural economics, financial literacy, saving behaviour and consumer debt. Articles were analysed using Biblioshiny (Bibliometric in R software) to evaluate publication growth, citation patterns, author productivity. It also helped find the journal impact, thematic evolution along with the collaboration networks.

Key Findings: Results show that research output surging from 44 publications in 2015 to 223 in 2025, it shows the rising global interest. The United States leads in both the publication volume and citations followed by other countries that are emerging as strong contributors. A few prolific authors and institutions dominate the field while the newer contributions add variety. Thematic analysis shows that nudging, financial literacy, saving behaviour and consumer psychology add a new dimension to the field.

Originality: This study provides the first comprehensive bibliometric overview of behavioural economics and financial decision making. It maps intellectual structures, global collaborations and thematic trends. It offers valuable insights for researchers and policymakers to guide future work in improving the financial decisions that can be taken by the people.

Keywords: *Behavioural Economics, savings behaviour, Consumer debt, Financial Literacy, Bibliometric analysis*

1. Introduction

Behavioral economics is based on human behaviour. They are subject to emotions and spontaneity, and who are swayed by their surroundings and circumstances. It tells that people do not always make what neoclassical economists consider the “rational” or “optimal” decision, even if they have all the information and tools to do so. Whereas traditional economics models tell us that they treat individuals as solely rational actors who have perfect self control and never lose sight of their long term goals or as people who occasionally make random decisions that cancel out in the long run.

Behavioral economics, though formally emerged in the late 20th century, has intellectual roots in Adam Smith, who recognized human tendencies such as overconfidence, loss aversion, and short-term bias. These insights were later expanded by scholars like Daniel Kahneman, Amos Tversky, and Richard Thaler, who is widely regarded as a founder of the field. Thaler’s work, which earned him the 2017 Nobel Prize in Economic Sciences, demonstrated predictable ways in which real human behavior deviates from classical models—for instance, through the sunk cost fallacy, where people persist in commitments they have invested in, regardless of risk. He also popularized the concept of the “nudge,” showing how subtle shifts in choice architecture can guide better decision-making, often influenced by factors such as mental accounting. The nudge in behavioral economics that is suggested by Richard Thaler’s is that it is a way to manipulate people’s choices which leads them to make specific decisions. Richard Thaler's ideas about the nudges got popularised in *nudge: improving decisions about wealth, health and happiness* in his 2008 book. Former UChicago legal scholar Cass Sunstein, now of Harvard University, (behavioral economics, explained. Witynski, M. 2024)

Financial decision making refers to the ways through which governments, households, firms or individuals make decisions about how to manage, assign and utilize financial resources over a period of time. It includes options related to investments, spending and borrowing, while balancing present needs with future goals. In traditional economics, financial decisions have been assumed to be rational like people measure costs and benefits, analyze risks and choose the option that maximizes their wealth and utility. However, behavioral economics shows that financial decision making is often influenced by the psychological behavior of humans. For example individuals may undersave due to present bias and with overconfidence they take high risks or accumulate high interest rate debt due to impulsivity.

By understanding common biases, people can make more practical and rational choices regarding savings, budgeting, and investing, leading to improved financial well being. This would help them to make better financial decisions. Individuals would become more aware and could recognise their irrational emotions or cognitive shortcuts when they are making one. and this would lead to a realistic approach to their finances. While the principle of behavioral economics can help in giving personalized financial plans that relate with human tendencies, making them more effective and efficient. It helps in determining long term financial goals, retirement plannings, children education etc. also impacts ability to focus on emergencies like medical bills, job loss. For individuals and households poor financial decisions can hamper them in present difficult scenarios which would lead them to a loophole of stress, anxiety and vulnerability. (*SSRN Electronic Journal, Lyons and Kass-Hanna 2021*)

Financial institutions and companies can make products and strategies which align with how people actually behave. Using “Nudges” for good, policymakers can assist people to make better decisions like prompt tax payments, businesses can get a better understanding of social influence and scarcity to design more effective marketing campaigns that would attract consumer psychology and drive sales. Insights from behavioral economics can lead to more efficient public policies in areas like health and sustainability, as solutions are tailored to real world human behaviour rather than assuming rationality. In this way, the ability to make good financial decisions impacts the profit, growth and competitiveness in firms and businesses. It also gives a mindmap on how to make investment choices to survive in competitive markets. (*Investopedia, kenton, 2024*)

Behavioral economics has emerged as a crucial field that bridges the gap between psychology and economics, offering valuable insights into how individuals and households make financial decisions. This perspective is important because in real life when financial decisions such as savings, investments, borrowing, managing debt come into the scenario we rarely make decisions on the perfect information or with complete rationality instead people tends fluctuate from standard models which directly affects their financial well being and economic outcomes. Given the increasing significance of this field, it is valuable to find out how research on behavioral economics and financial decision making has evolved over time. Conducting a bibliometric review allows for a systematic mapping of global research trends, identifying influential authors, highly cited publications, prominent countries and emerging themes. By doing this we can find out possible outcomes on our research questions such as

- i) how has the volume of global research on behavioral economics and financial decision making evolved over time?
- ii) who are the influential authors, institutions and countries contributing to this field?
- iii) how have they shifted over different time periods?
- iv) which models, theories or concepts from the intellectual base of behavioral economics research in the context of financial decision making?

2. Research Methodology

The topic itself is not a novel topic; it has been researched over many years. In recent years study on this topic has shown significant growth, here arises the need of doing bibliometric analysis of this topic. Research on behavioral economics and financial decision making is critical because it challenges the traditional assumption of rational actors. Bibliometric analysis handles a large amount of data making identification of trends and themes and major contributions convenient.

Boolean search was used on Web of Science(WoS) database to fetch relevant literature published between 2015-2025. The research team used prior understanding of the subject and published researchers to determine the keywords for the present study. The research parameters include “behavioral economics” AND “household finance” OR “saving behavior” OR “consumer debt” OR “financial literacy”. A period of ten years was chosen to strictly focus on the recent development of research. The documents were extracted from the database, and were imported in Biblioshiny(Web interface for Bibliometrix analysis).

For the sake of estimation subtleties, only articles and reviews published in the English language were examined, considering the most reliable scholarly contributions to the knowledge base under investigation, and the results were sorted by citation count, resulting in 4106 samples.

Precisely, this paper analyses the following indicators:

- i. Overview: main information, annual scientific production, average citations per year and three field plots.
- ii. Sources: Most relevant sources, Bradford’s Law, sources local impact and sources’ production over time.

iii. Authors: Most relevant authors and affiliations, authors and affiliations' production over time, Lotka's Law, impact, countries of corresponding authors', countries' scientific production, country's production over time and most cited countries.

iv.Documents: Most global cited documents Words: most frequent words, word cloud, tree map, words' frequency over time and trend topics.

v.Clustering by Coupling.

vi.Conceptual structure: co-occurrence network, thematic map, thematic evolution and factorial analysis.

vii.Social Structure: Collaboration network, Countries collaboration world map.

2.1 Results

Table 2.1 is a summary of the Bibliometric metadata. The published record comprised 2770 authors and 1000 documents rising from 44 in 2015 to 223 in 2025.

Description	Results
MAIN INFORMATION ABOUT DATA	
Timespan	2015:2025
Sources (Journals, Books, etc)	669
Documents	1000
Annual Growth Rate %	17.62
Document Average Age	3.86
Average citations per doc	7.301
References	0
DOCUMENT CONTENTS	
Keywords Plus (ID)	1112
Author's Keywords (DE)	2581
AUTHORS	
Authors	2770
Authors of single-authored docs	152
AUTHORS COLLABORATION	
Single-authored docs	177
Co-Authors per Doc	3.14
International co-authorships %	18.8
DOCUMENT TYPES	
article; book chapter	76
article; early access	272
article; early access; retracted publication	1
article; proceedings paper	19
editorial material	25
editorial material; book chapter	10
letter; early access	1
proceedings paper	403
review	185
review; book chapter	1
review; early access	7

Table 2.1

3. Bibliometric Review

3.1 Overview

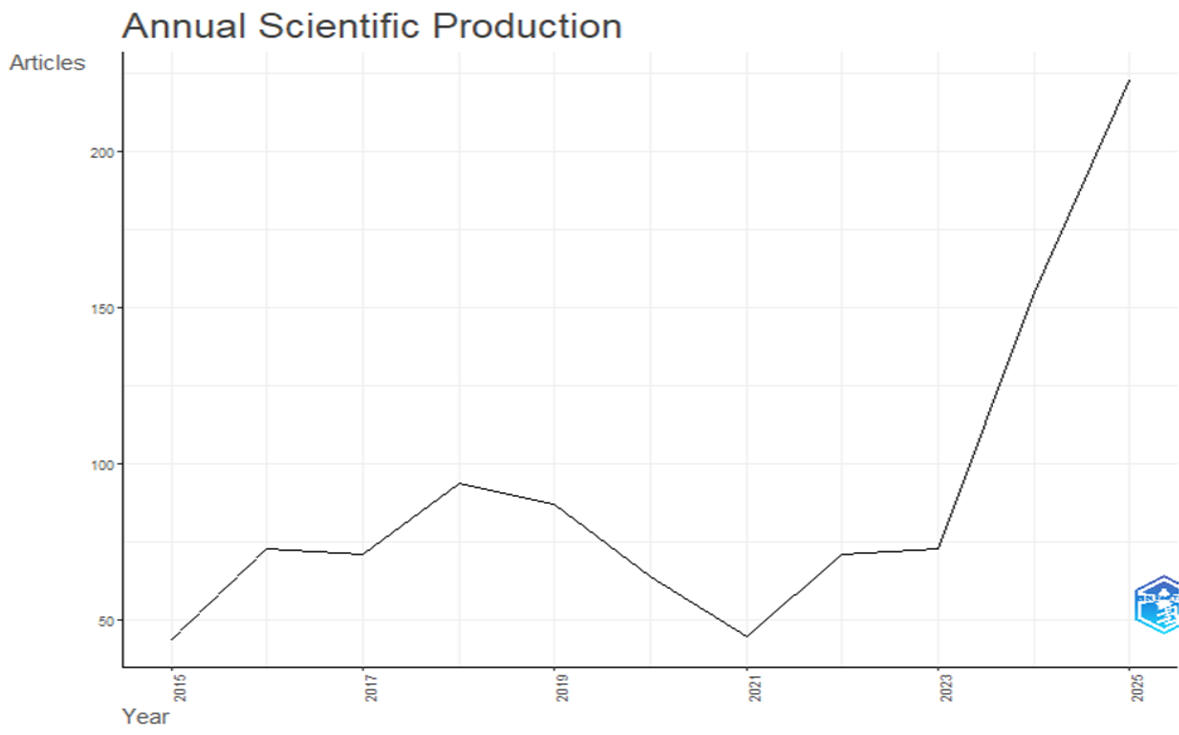
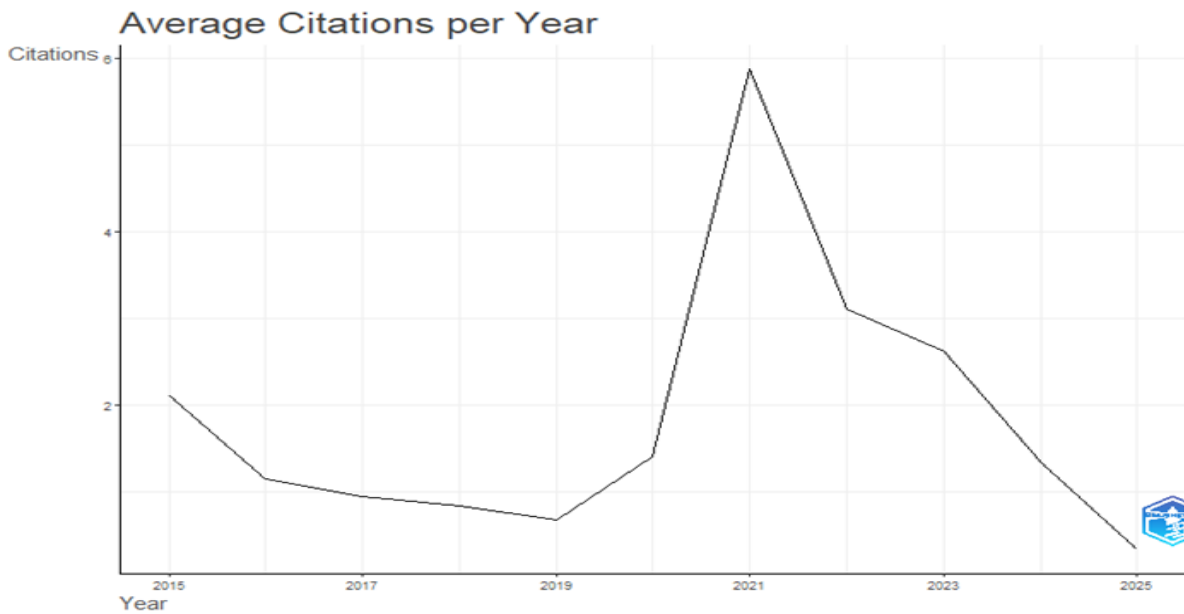
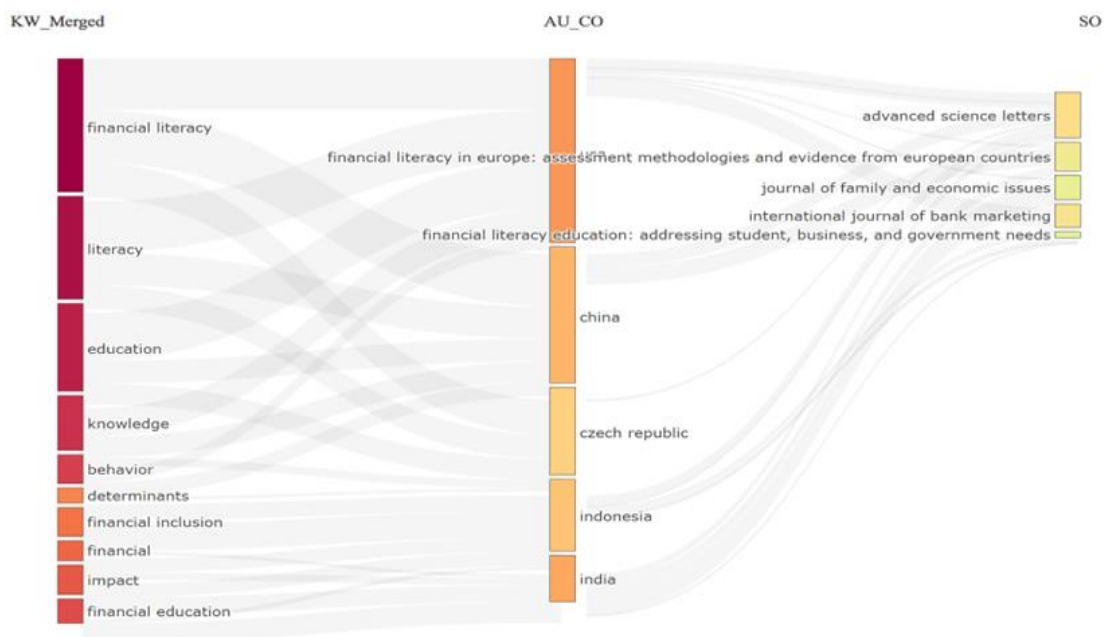


Fig 1 Annual scientific production of articles

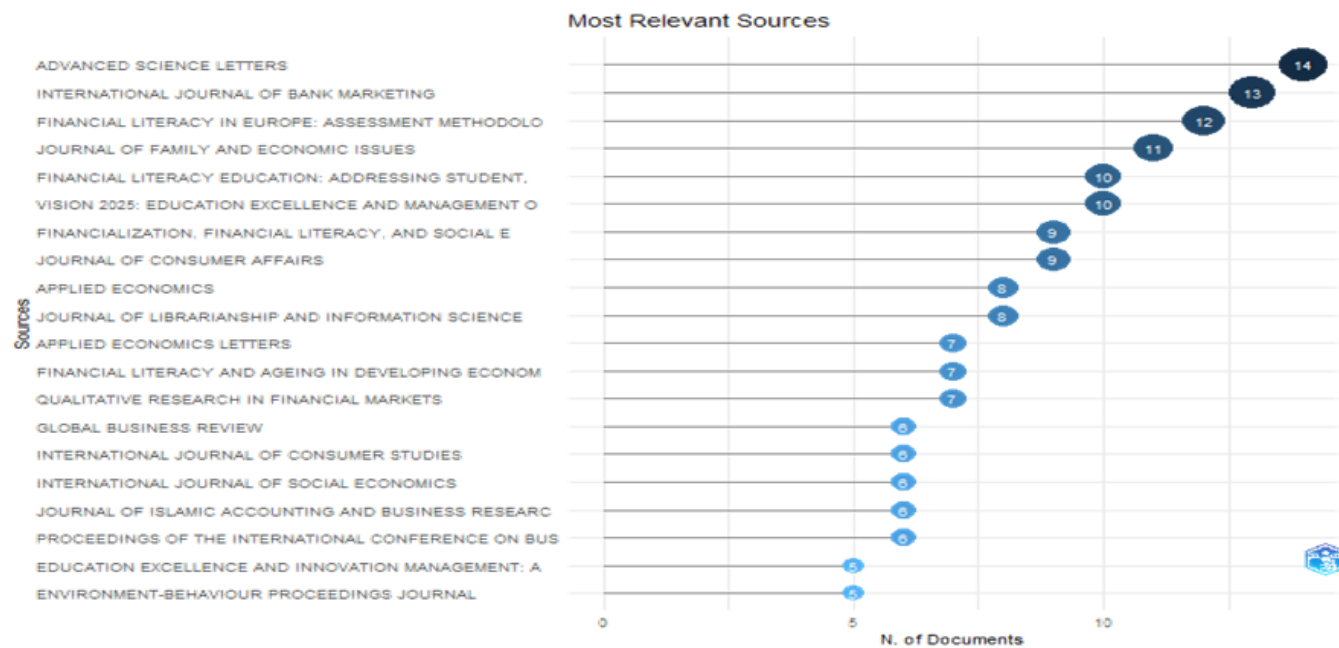
This figure 1 shows the trendline of studies that took place over the last decade. In the beginning of the decade the production was about 40-45 articles which rose to 75 articles in the next year. After a slight dip in 2017 articles rose to around 95 in 2018. The production dropped to its lowest in 2021, COVID-19 pandemic could have affected the research. After 2021, the production gained speed and started to rise at an increasing rate. In 2022 the article production was around 75. After 2023, the production was highest in the period of 2023-25. In 2025 production reached the highest level i.e 225 articles. Although this study has been conducted before the end of 2025, the peak shows that there are increased chances of more production till end of this year and next year.



The period of 2020-2021 had the lowest production but highest citations. This indicates that the papers published were prominent and had a huge impact on other researchers. In the period of 2023-2025 where the production increased immensely the impact of those studies was minimal. This indicates the papers published recently do not have great influence on other studies.

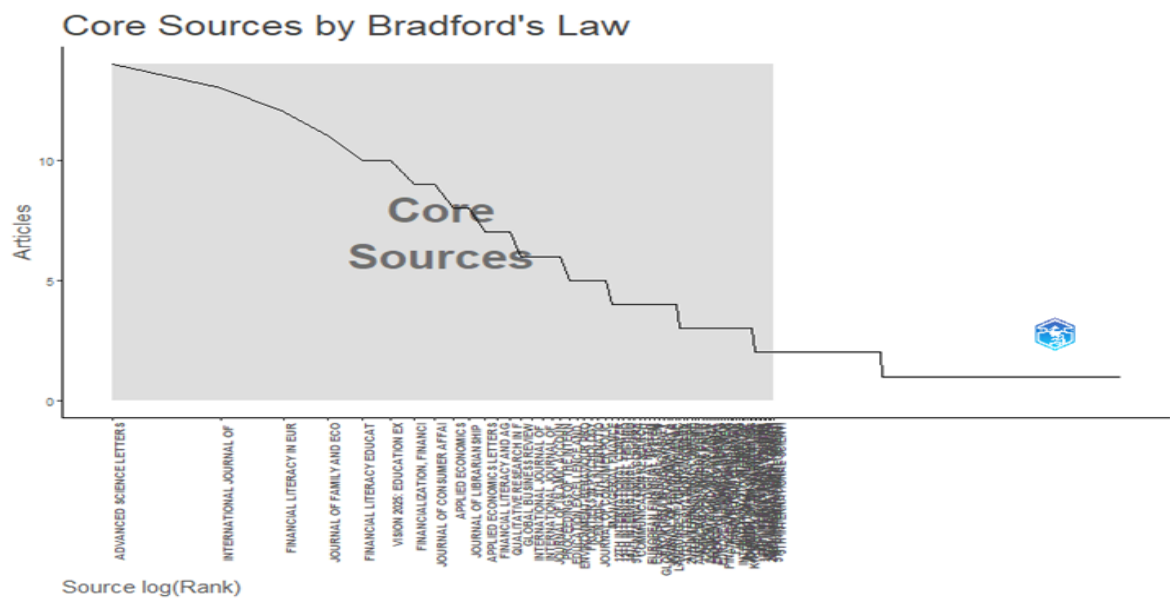


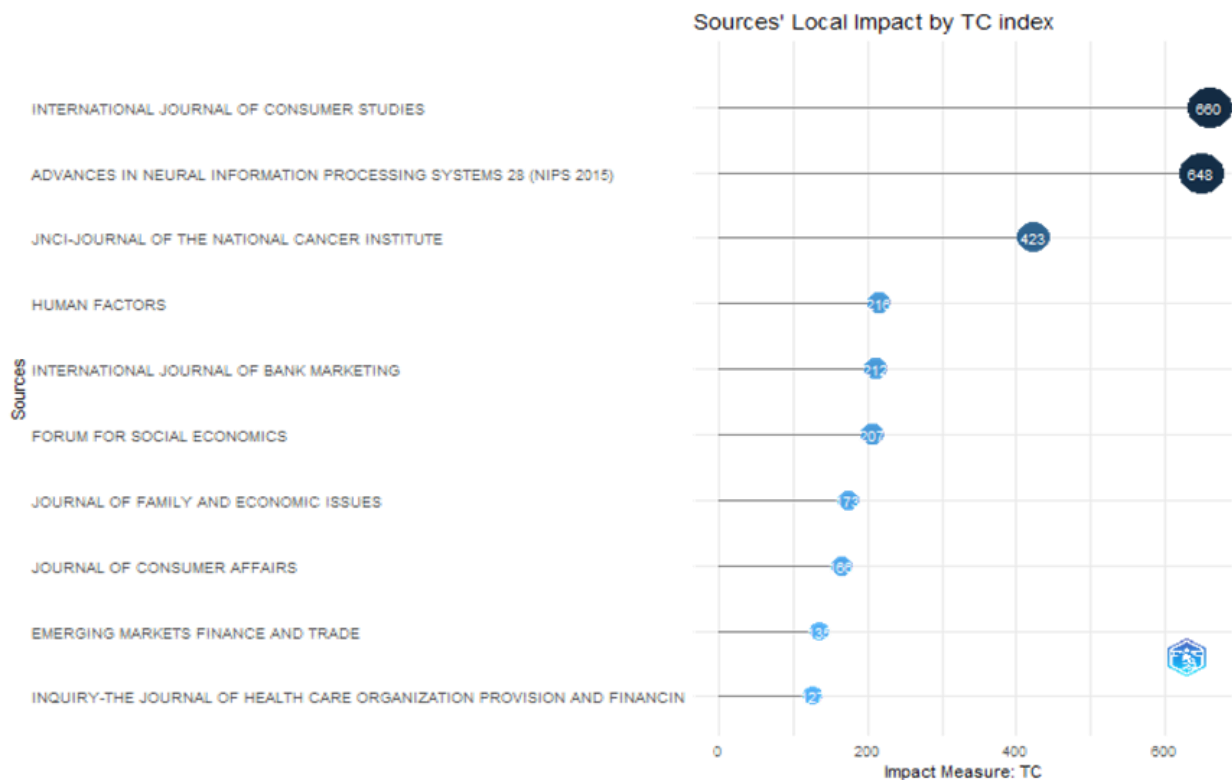
3.2 Sources



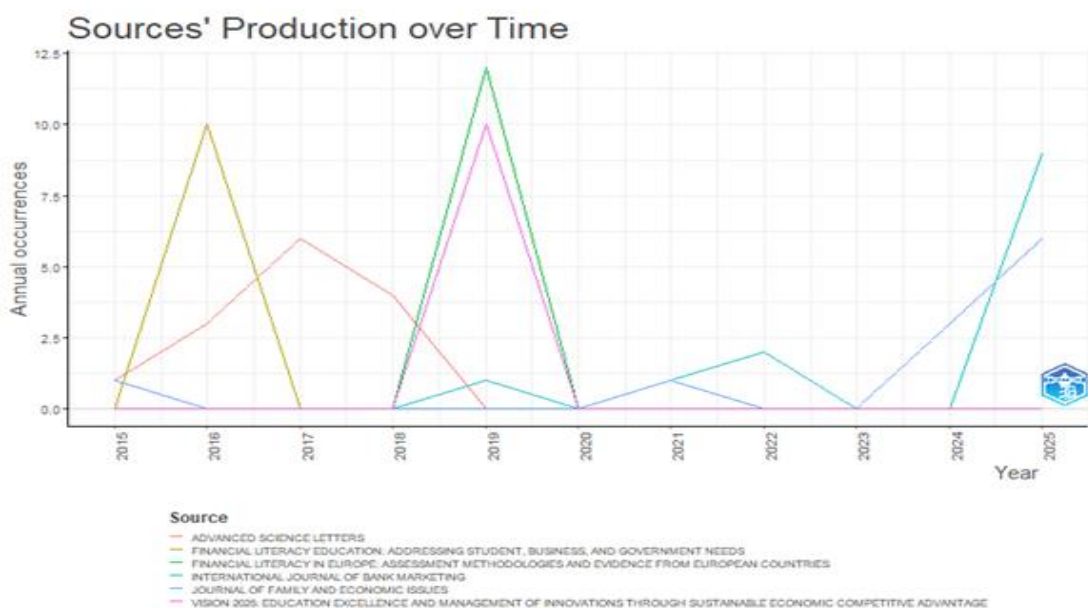
3.2.1 Bradford’s law

Bradford's law proposed by Samuel C. Bradford in 1934, a bibliometric principle used to identify the “core journal” (or sources) that publish the significant portion of the research topic. The law states that if journals are arranged in order of decreasing productivity they can be divided into core zones and subsequent zones of decreasing relevance.





This above graph shows that a few sources, International journal of consumer studies and advances in neural information processing systems, have the highest citation impact, while other sources contribute fewer citations-illustrates Bradford's law of concentrated of influential sources.



3.3 Authors

3.3.1 Most Relevant Authors

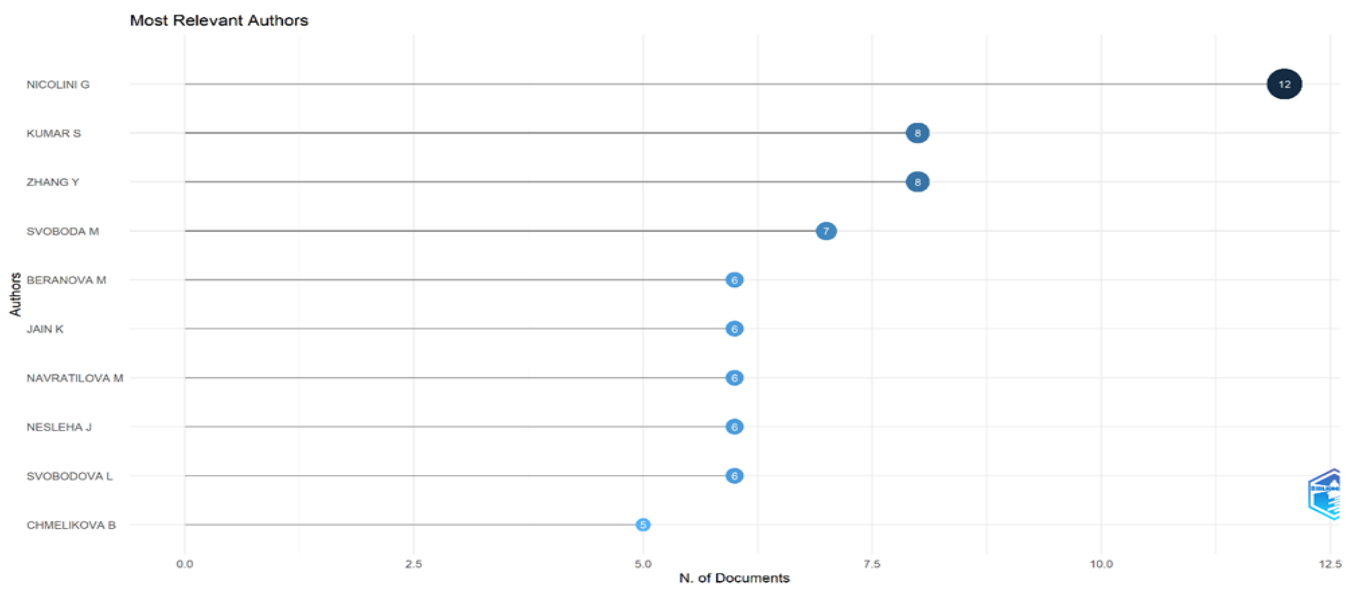
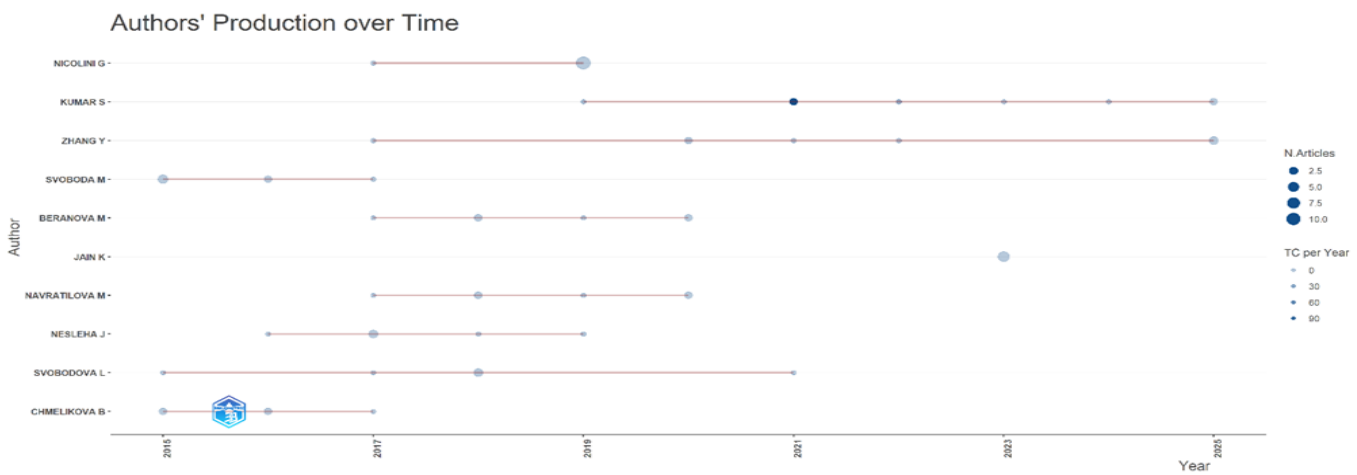


Fig 2 most relevant authors

The analysis of the most relevant authors reveals that a limited number of scholars dominate the publication, reflecting their central role in shaping discourse. These authors contribute substantially to advancing research directions. The high publication output signifies not only productivity but also recognition within the academic community. The concentration of emerging authors remains lower leaving room for diversification and broader participation.

3.3.2 Author's Production over Time



The trend of author’s production over time illustrates how individual contributions have evolved. It reflects both the continuity and the entry of new scholars. The pattern shows that a few authors have maintained consistent output over several years. The short-term contributions suggest exploratory interest and diversification of research focus. Such dynamics highlight that a core group of authors drives ongoing research creation. There are also new authors as well who have periodic inputs. This combination ensures the stability and innovation in scholarly landscape and supports the thematic expansion over time.

3.3.3 Lotka’s law

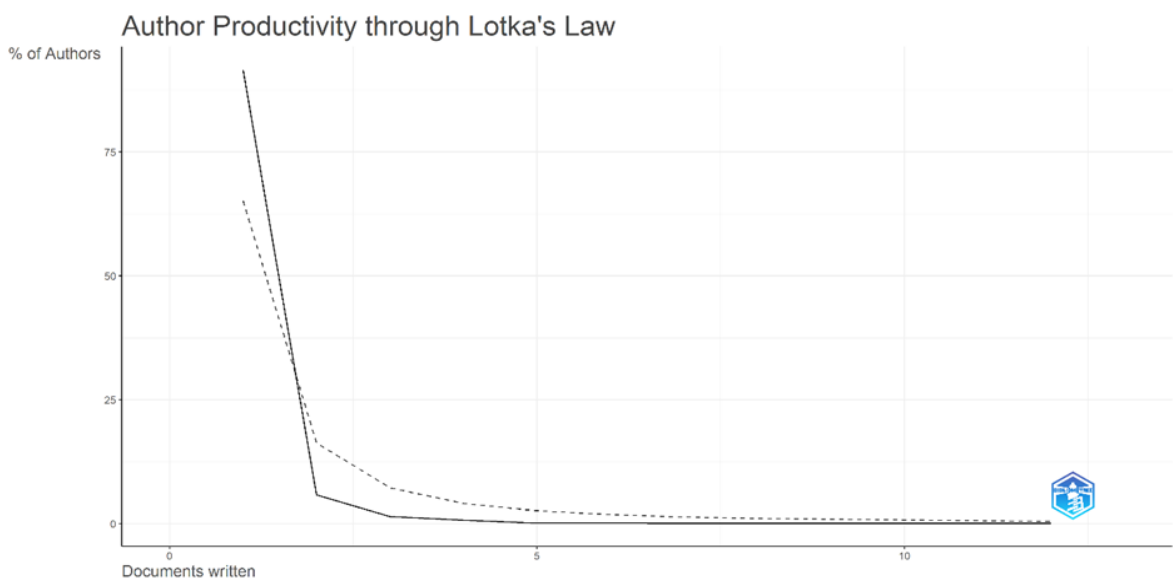


Fig 3 Author productivity through Lotka’s law’

The Lotka’s Law shows that only a small fraction of authors produce the majority of publications, while most contribute only twice or once. The pattern reflects the principle of scientific productivity where the research output is highly skewed towards the few prolific contributors. The dominance of these leading authors underscores their central role in shaping the field’s evolution and building theoretical base. On the other hand, the large number of intermittent contributors indicates mixed academic interest, with numerous researchers investigating the subject without intense focus. This kind of spread indicates both robust leadership in the development of knowledge and space for broader participation to counterbalance the density of contributions.

3.3.4 Author’s local impact

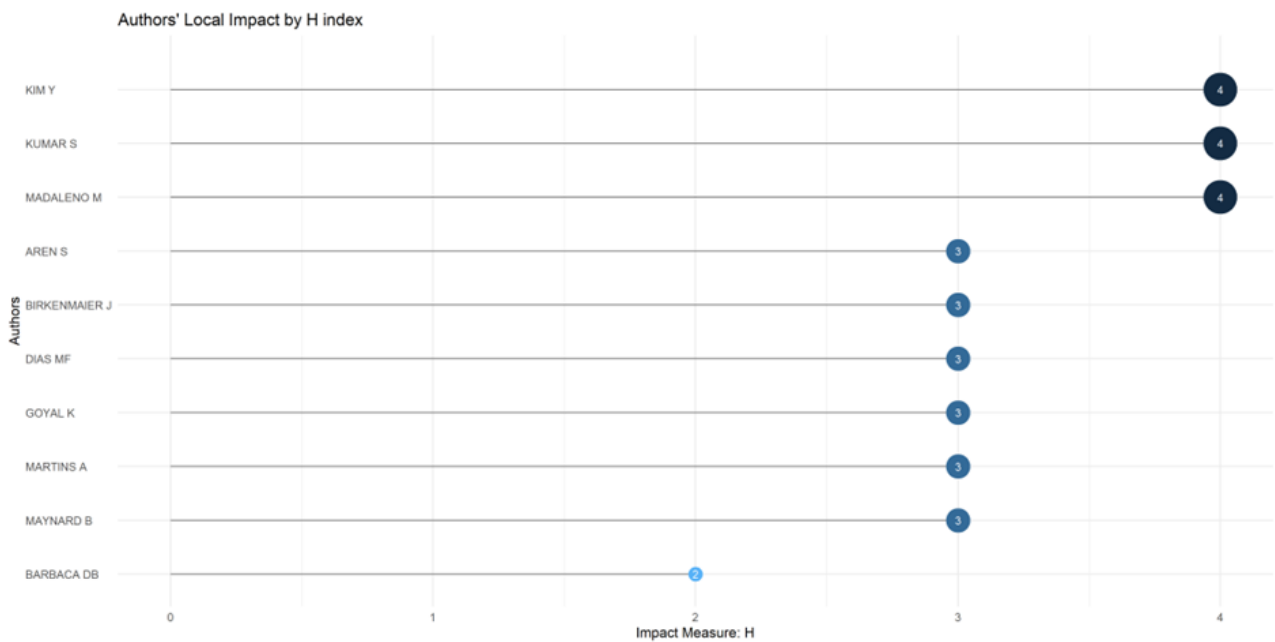
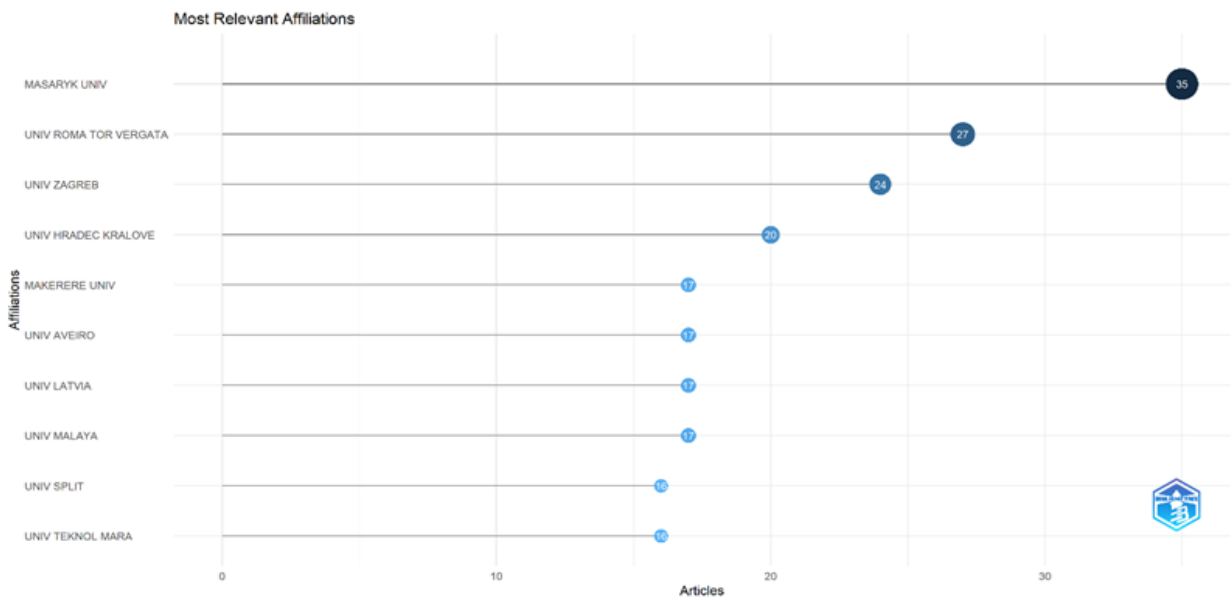


Fig 4 authors’ local impact

The study of authors' local influence highlights influence quantified not only in terms of productivity but also in terms of relevance in the dataset. Some authors stand out as having high local citation counts, suggesting that they are strongly influential with respect to peers within the same field of research. This implies that their research contributes fundamental knowledge or popular methodologies, rendering them extremely referential in informing future scholarship. In contrast to international citation patterns, local influence emphasizes contextual significance, revealing how particular work informs the thematic evolution of the discipline. This acknowledgment certifies the decisive role of leading authors in fixing the intellectual framework of the research domain.

3.4 Affiliations



3.4.1 Affiliations' Production over Time

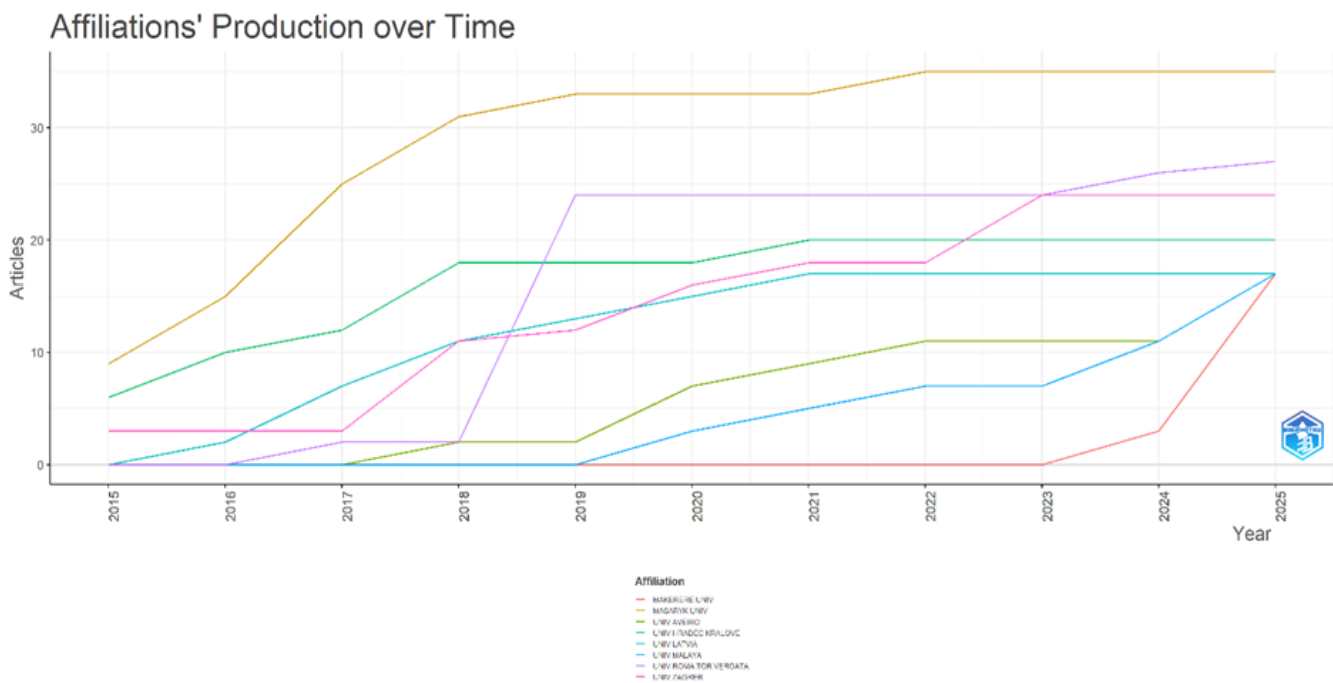
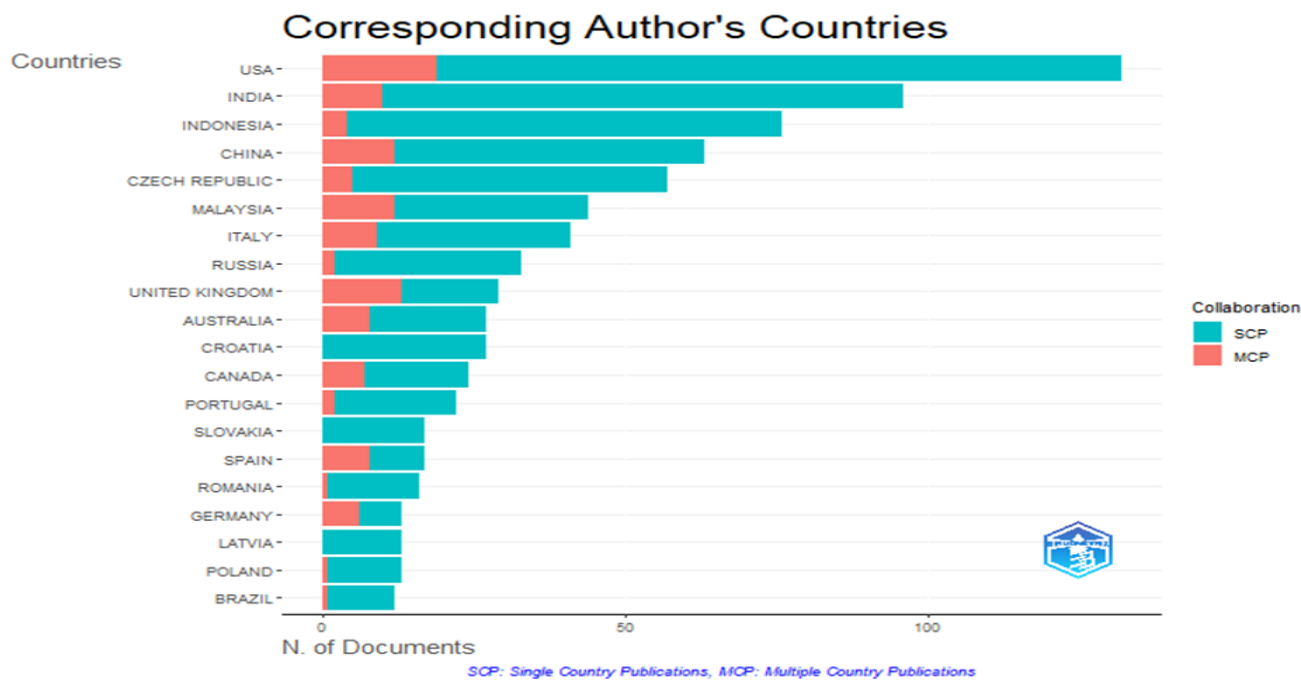


Fig 5 affiliations production over time

The trend analysis of affiliations' output over years reveals that research activity is highly skewed among a few top institutions that consistently make contributions year after year, and many others exhibit occasional activity. The trend line reveals steady growth with sharp peaks, which implies that institutional production is balanced by collaborative work, funding patterns, or thematic research calls. Some universities have a consistent line, demonstrating entrenched skill and recognized research teams in the field. By contrast, sporadic blasts from other institutions signify exploratory participation but not long-standing specialization. Such unbalanced distribution supports the influence of core institutions in defining the research area, yet at the same time indicates potential for newer universities to reinforce their position through ongoing input.

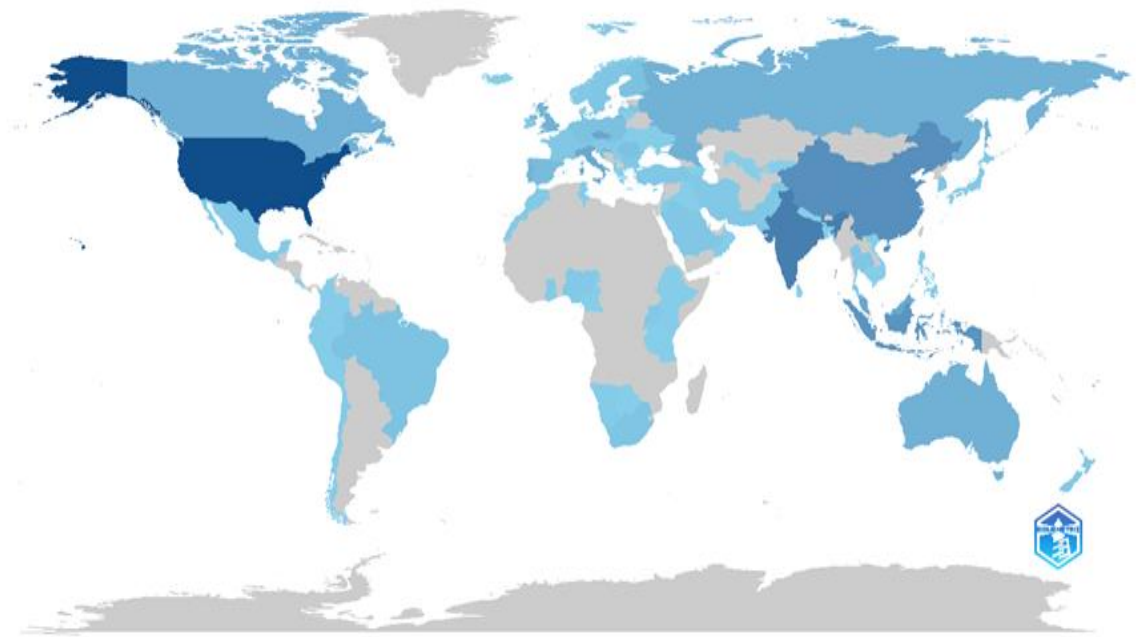
3.5 Countries

United States is the top corresponding author's country. This is due to the robust academic foundations and wide-reaching research networks in the field. The USA dominance can be traced to established institutions, greater research funding, and global collaborations that further accentuate its profile. India and China rank second and third, respectively, with both reporting robust growth as emerging research centres, mainly attributed to growing investments and their enormous-sized applicability in the field. Shown in the below graph.

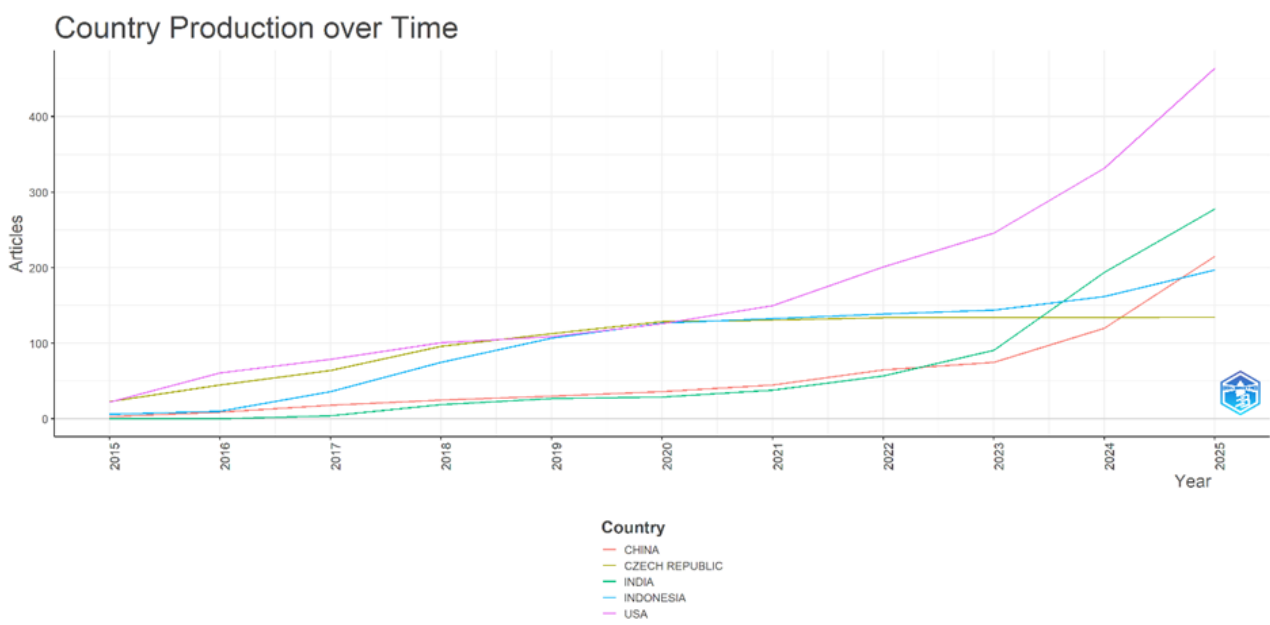


3.5.1 Countries' Scientific Production

Country Scientific Production

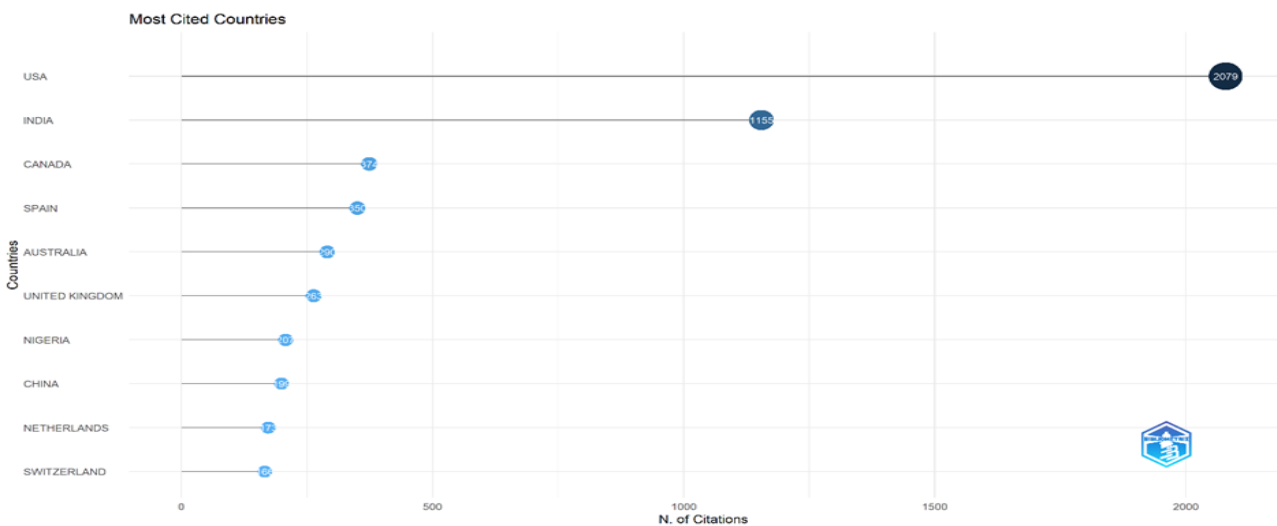


3.5.2 Countries' Scientific Production Over Time

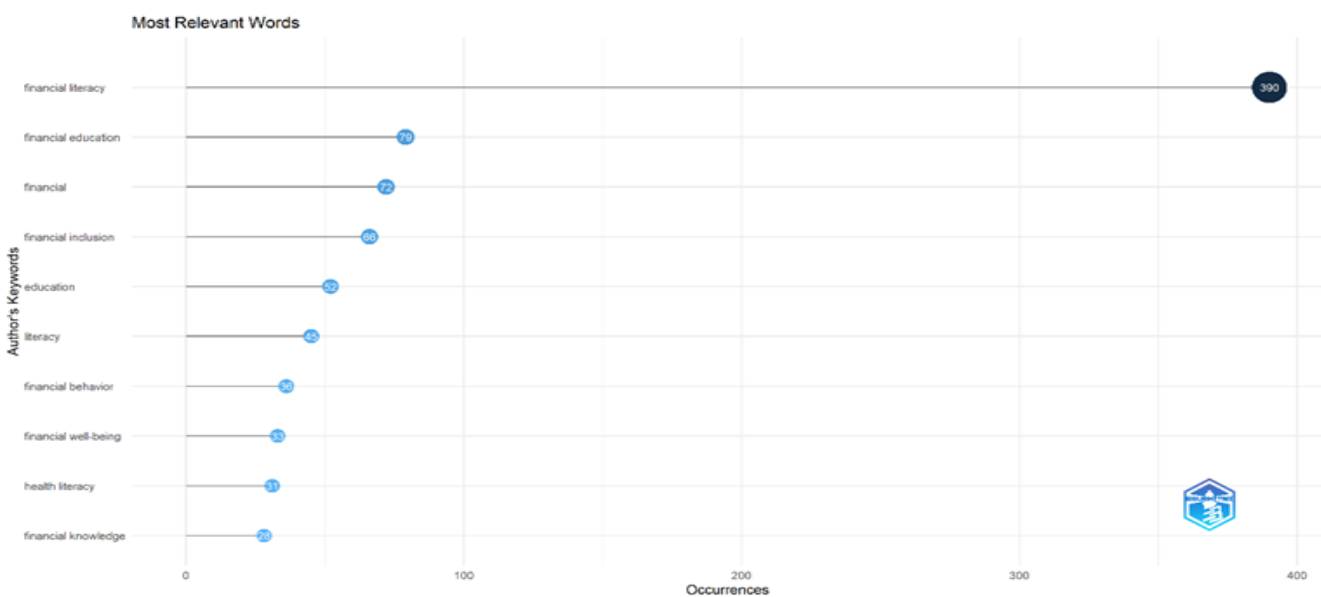


The trend shows how the United States has consistently led the research output maintaining dominance over years. India and China show upward growth with recent years marking sharp increase in publications and rising interests and investments and academic interest. Indonesia and the Czech Republic emerge with modest contributions reflecting early but growing interest in the field. Their gradual entry highlights the expanding geographical scope of research and opportunities for wider international collaborations.

3.5.3 Most cited countries



3.6 Analysis of Documents



The above figure shows that the research paper has the most frequently used keywords and those keywords are financial literacy, financial education, financial inclusion, financial behaviour, financial well-being,

health literacy, and financial knowledge.



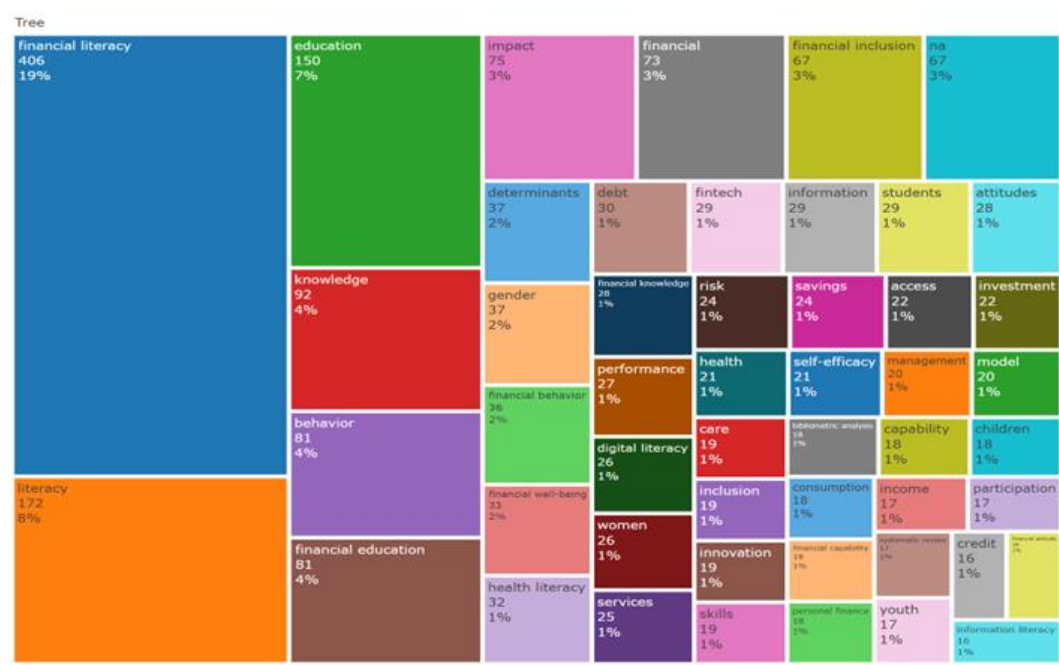
Fig 6 wordcloud for keyword plus

This figure 6 shows the wordcloud for keyword plus. The main terms are education, literacy, knowledge, and behaviour highlight the focus of the research- to explore how knowledge mainly gained through education results in financial actions. The second important term is “impact” showing that literature is focusing towards results and is dedicated to analysing the success of intervention. Moreover, the smaller term networks bring out the field’s necessary subdivision and its depth across different disciplines. It includes key demographic factors such as “gender differences”, “students” and “income” along with psychological factors like “self-efficiency” and “attitude”.



Fig 7 wordcloud for authors keywords

The above graph 7 shows the wordcloud for author’s keywords. Financial literacy concept is the most researched and crucial area that is used by the author. The other key practical such as “financial education” which helps to gain skills, “financial behaviour” and “financial well-being” which represents the result and “financial inclusion” which provides social context. This research area mainly concentrate on turning the financial knowledge into a positive outcome in real world.



The above figure shows the treemap of top 50 keywords in the data set. The most research works involves financial literacy, financial education, financial inclusion, knowledge, and behaviour. It also examines the additional factors like gender, students, risk, and self-efficiency which help the research to classify and explore psychological mechanism.

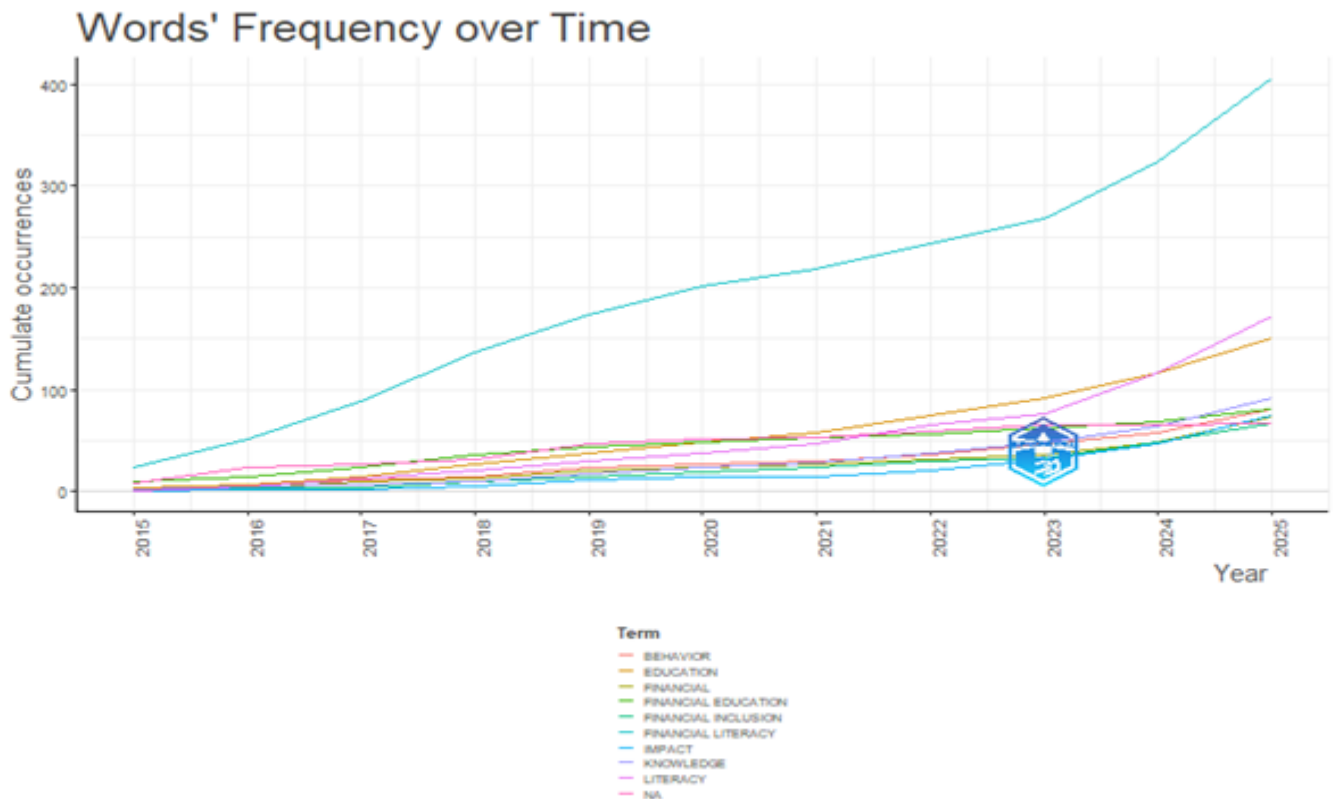


Fig 8 words frequency over time

The figure 8 shows the cumulative occurrences of words over time. With time, Financial literacy has increased significantly from 2015 to 2025. It is also shown that education, financial, knowledge, literacy, financial education, financial inclusion, impact has also increased. This means researchers are more focused towards the practical side. They want to know what specific lessons are effective, how those lessons will change people's actions and how to make sure everyone has fair access to the financial tools.

the trend research topics from 2016 to 2024. The two factors that researchers are always focused on are "financial literacy" and "financial behaviour". However from 2023 the recent research reacted to real world problems- big words like "Covid-19" and "trust" suddenly became popular. This means now the researchers are focused on "how the crisis" and "people's trust" on banks affects their money decisions. Meanwhile, topics like "impact", "inclusion" and "education" are also increasing rapidly showing that researchers are now even more than interested on finding out which programs works and benefit all.

intellectual overlap. The presence of multiple data related to 2025 highlights the recent surge in academic output and growing interest and activity in this research area.

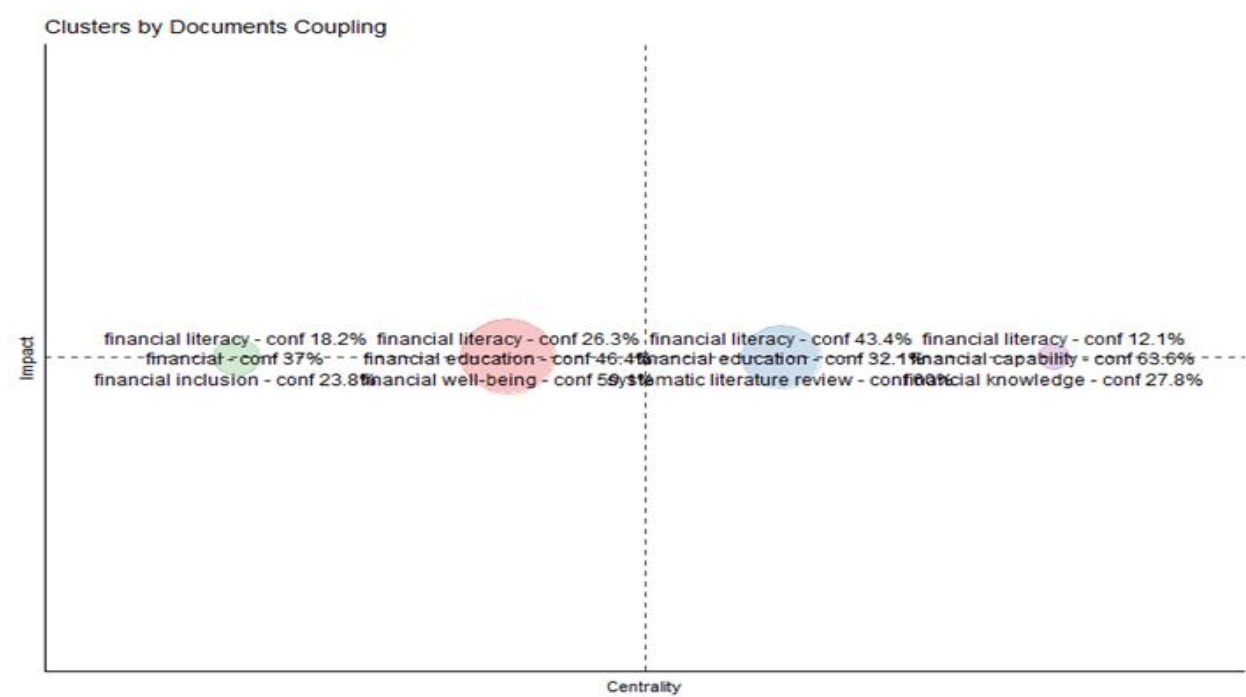


Fig 9 clusters by documents coupling

The figure 9 shows the clustering of documents based on coupling, categorized by thematic focus such as financial literacy, education and capability. The x axis represents centrality, indicating how connected each cluster is within the network while y axis shows impact. Most clusters lie close to the center indicating moderate centrality and impact. The “ financial capability – conf 63.6% “cluster stands out with high centrality suggesting strong connectivity in the literature. Clusters such as “ financial literacy – conf 43.4%” and “financial education – conf 46.4%” are prominent. This indicates growing research interest in financial themes with overlapping domains.

3.8 Conceptual Structure

This section of network analysis will help the readers to understand the occurrence of all keywords,author keywords,abstracts,and keywords plus .The thematic network will help to analyse the thematic map and thematic evolution. To run the co occurrence ,the edge size has been kept at 5 and node side at 3. Figure 10 shows the co occurrence of all keywords.The research papers have produced a prominent network focusing on mainly three clusters.The first cluster of networks focuses on financial literacy,knowledge and literacy.The second cluster emphasises on financial inclusion,performance and access.The third cluster

focuses on consumption, debt and credit. This is because these represent the critical dimensions which strongly influence how households save, borrow and invest. Apart from these clusters it shows that behavioural economics and financial decision making is having different applications in other domains including health literacy, care, skills and digital literacy.

Fig 10 Co occurrence of all Keywords

Fig 11 Co occurrence of Author Keywords



Fig 12 :Co occurrence of Keywords Plus

Figure 11 shows the graphical analysis of co occurrence of author keywords. This will help the readers analyse what the researchers main focus has been while writing on these research papers. This can be divided into network of several clusters. The main primary cluster focuses on financial literacy, financial planning, financial management and education. The second cluster focuses on financial capability, financial inclusion and financial capability. The third cluster focuses on digital financial literacy, financial attitude and financial knowledge. There are other small clusters as well with less emphasis on higher education, scoping review, Covid 19, health literacy, consumption and debt. Figure 12 shows the co occurrence of keywords plus.

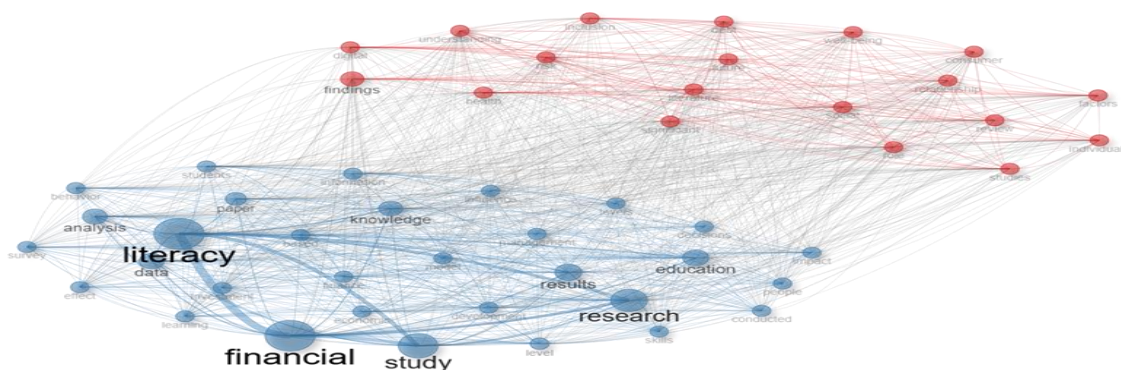


Fig 13 Co-occurrence of abstracts

Figure 13 shows the co-occurrence of abstracts. The bottom primary blue cluster highlights literacy, financial, study, research, analysis and results. This shows how important are education and

awareness in making a financial decision. It also reflects the knowledge driven side of behavioural economics. The Top red cluster focuses on significant , findings, review ,risk ,future and people. This reflects the importance of having significant outcomes and results.

Figure 14 shows the Thematic Map of Author Keywords. The thematic map plot is done on a bi dimensional matrix ,whose x axis represents Centrality (or relevance degree) and y axis represents the Density (or development degree). For this plot, the minimum cluster selected is 5 with 150 words. Centrality shows the importance of a theme in a broader research field and density reflects the development of a theme in that area. The upper right quadrant is the motor theme having high centrality and high density. Keywords like education, finance and knowledge forms one cluster under this theme, While health literacy, scoping review and mental health are grouped in another cluster. The upper left quadrant are the niche themes having high density and low centrality. We can find keywords like information literacy ,higher education ,learning, consumer credit ,household debt, Islamic financial literacy and risk being part of this theme. The keywords like debt ,consumption ,household finance in one cluster and financial literacy, financial education and financial in another cluster have high centrality and low density . These fall under the category of basic themes. These are the underdeveloped areas which has scope for further theoretical and empirical development . Then we have the emerging or declining themes having low density and low centrality . This includes keywords like digital literacy, literature review and media literacy. These areas are either newly growing or are losing significance. Figure 15 shows the thematic evolution of keywords from 2015-2022 to 2023-2025.

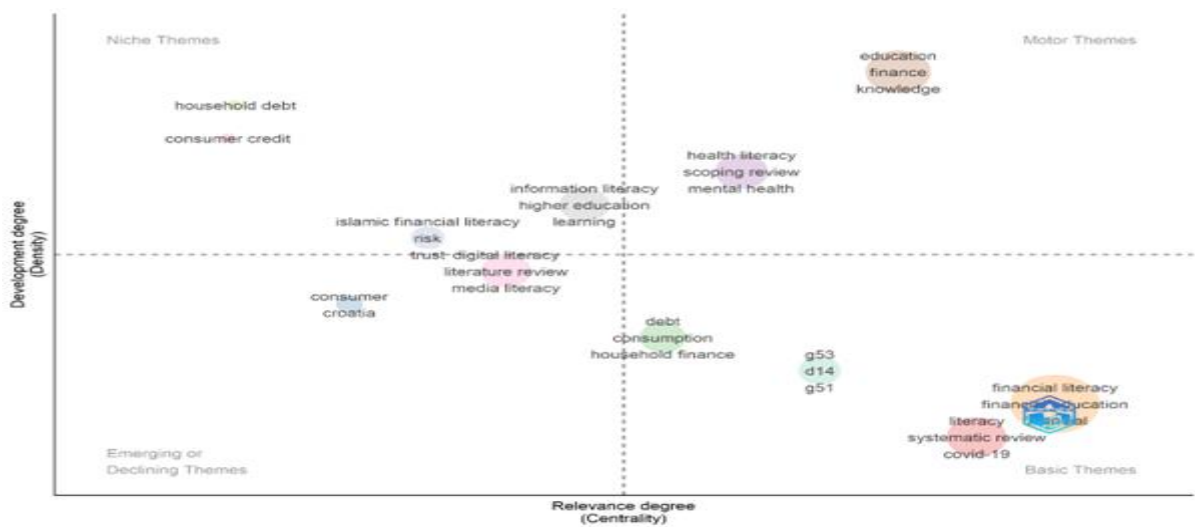


Fig 14 Thematic Map of Author Keywords

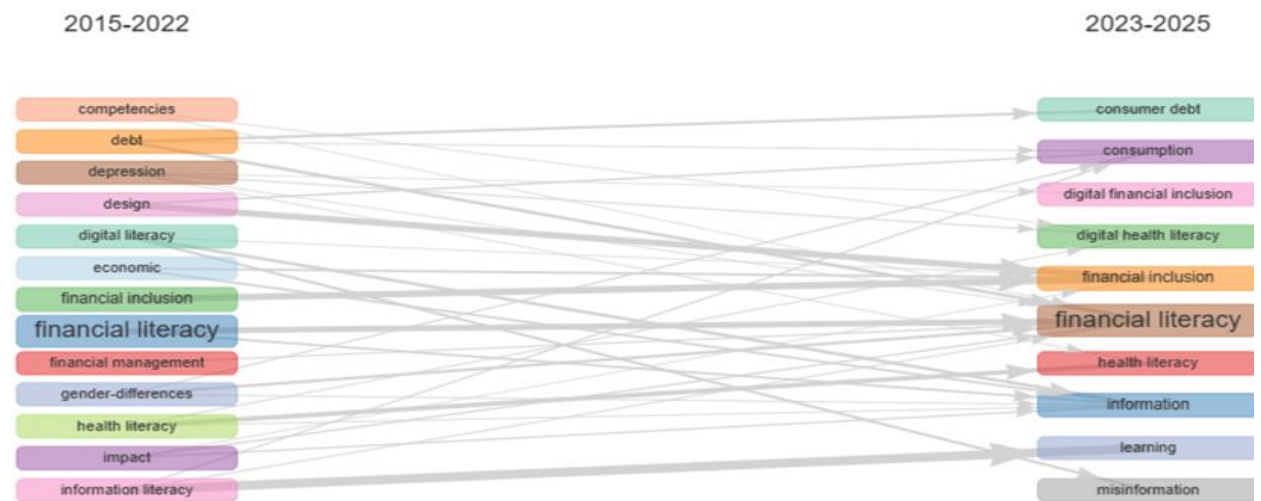


Fig 15 Thematic Evolution of Keywords

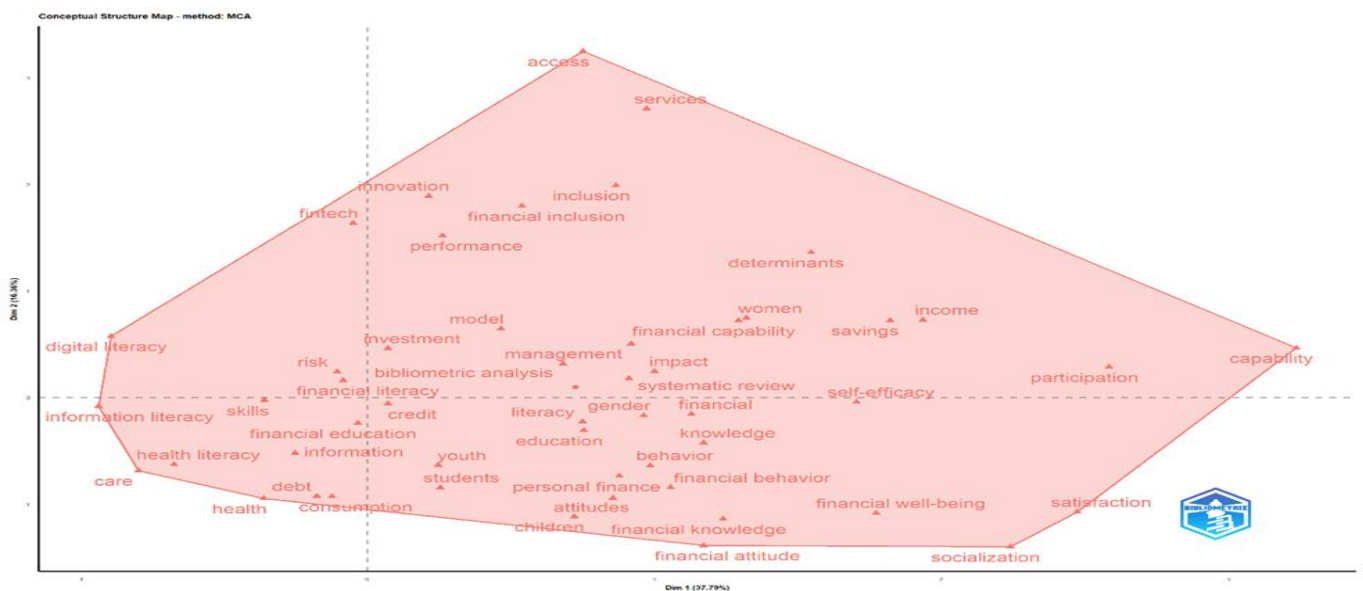


Fig 16 Word Map of all Keywords

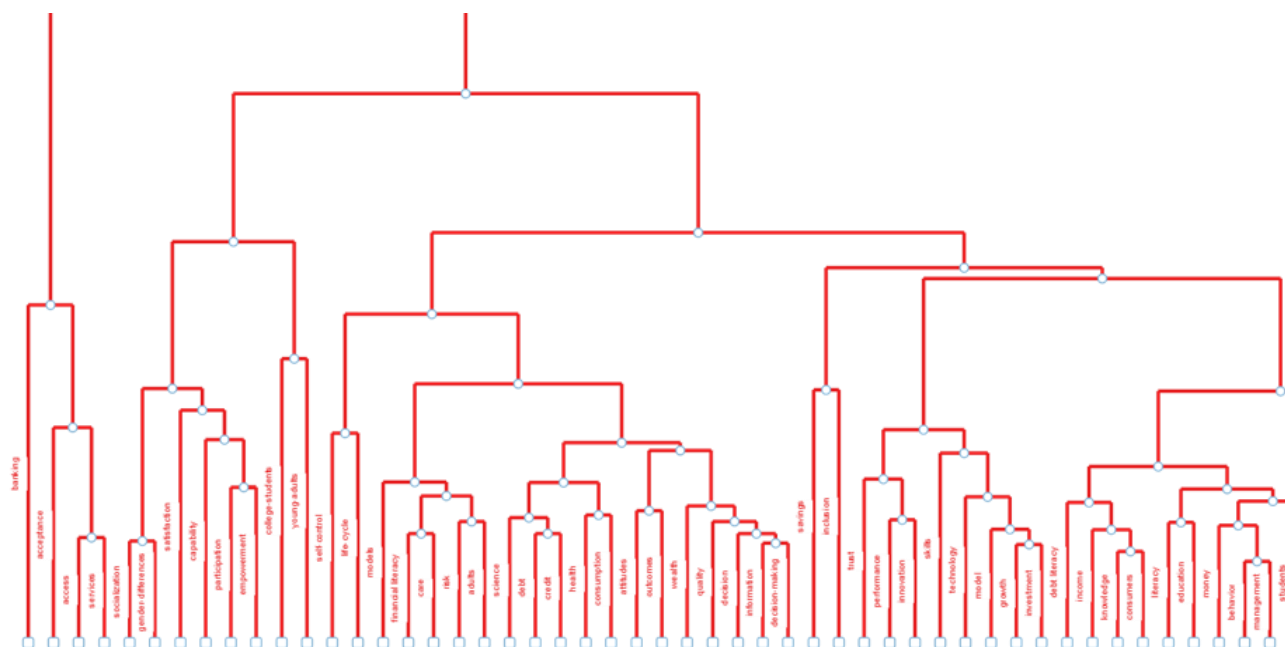


Fig 17 : Dendrogram of Keyword Plus

Figure 16 shows the word map of all keywords which is generated using the method of MCA(Multiple Correspondence Analysis).

It is structured around two dimensions Dim 1 and Dim 2 i.e 37.39% and 16.36 %.The horizontal axis (Dim 1)spreads from financial literacy ,financial education and skills on the left to capability ,participation,satisfaction and socialization on the right.While on the vertical axis(Dim 2)ranges over keywords like access,innovation,inclusion and performance on the top to more behavioural aspect like attitude,financial knowledge and financial well being on the bottom.The Conceptual structure map here reflects the multidimensional nature of our research area. .Figure 17 shows the dendrogram of Keywords plus.It reflects that the research forms cluster around major themes.Like one cluster focuses on education,knowledge,financial literacy while other cluster focuses on attitudes,skill ,bevaious and well being.Additional clusters focus on kewords such as innovation,investment ,access and inclusion.

3.9 Social Structure

Social structure shows how authors or institutions or countries interact with others in the field of scientific research.Figure 18 depicts the author's collaboration. While there are two prominent orange and green clusters showing influential authors, there are many small fragmented groups among different authors.This indicates that there is limited co authorship.Figure 19 shows the institutional collaboration network.Institutions in the red cluster stand out the most signifying that they are central to the collaboration and are likely the ones who publish the most.Many small institutions are also present connected with two

nodes reflecting weaker institutional linkages.

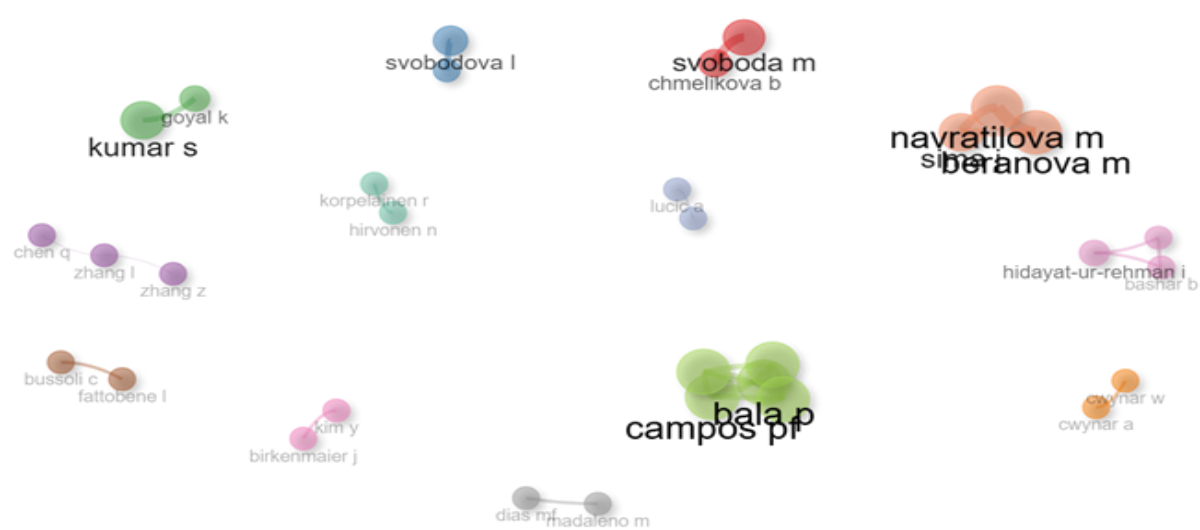


Fig 18 Author Collaboration



Fig 19 Institutions collaboration

Figure 20 shows the Country collaboration network. The blue cluster is the most prominent one indicating primary collaborations which includes USA, Indonesia, Malaysia, China, South Africa. It shows the cross country collaborations with Asian Countries. Purple and Green clusters form the secondary collaboration with countries like Austria, Germany, Poland, Australia, Brazil and Portugal. There are other peripheral nodes which indicates weak or less collaboration. It includes countries like Japan, Malta and Mexico and Turkey. Figure 21 shows the Country collaboration worldmap. The map shows that the research is dominated by strong ties between US, Europe and Asia. USA stands out as a central hub. This is due to the fact that USA is a global leader in behavioural economics and is backed by strong institutional support with Asian countries like India, China, Indonesia and Malaysia emerging as major collaborators. While some regions are still developing signalling the need for broader inclusion in global collaboration.

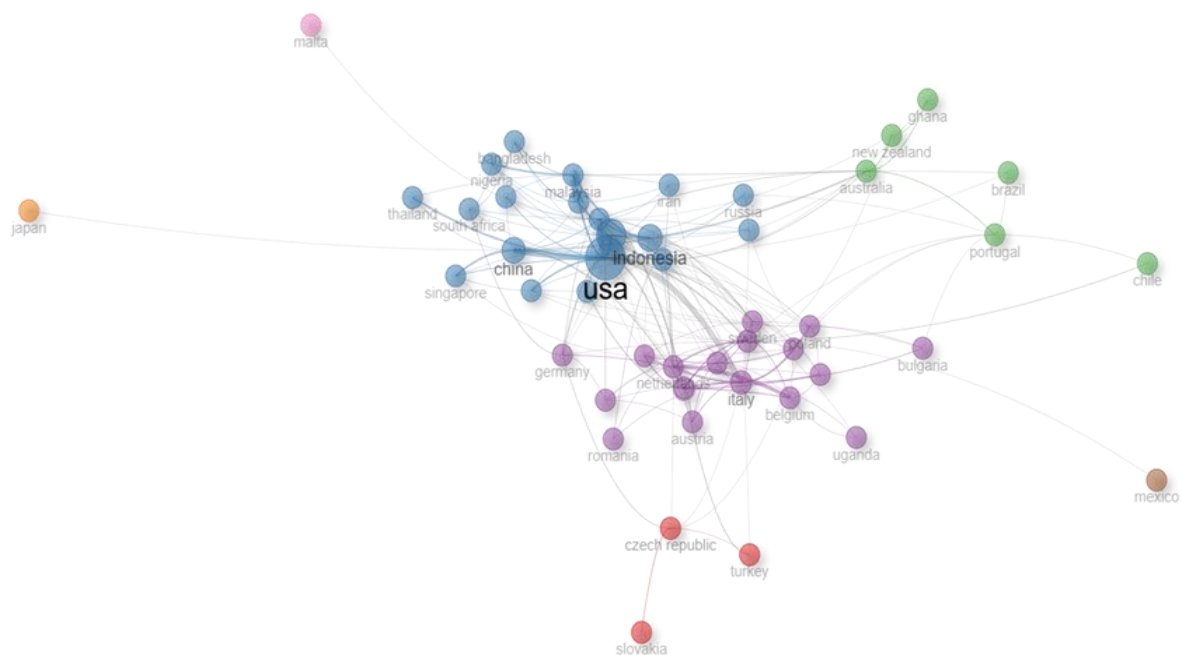


Fig 20 Country Collaboration

Country Collaboration Map

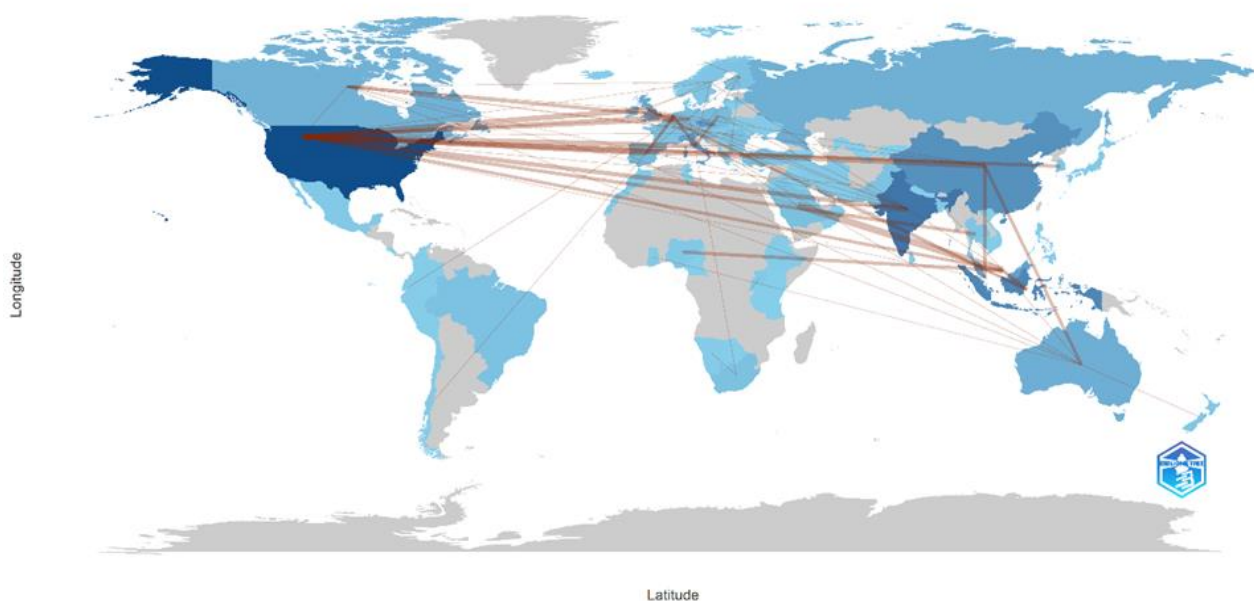


Fig 21 Country Collaboration Worldmap

4. Conclusion

This bibliometric analysis provides a quantitative overview of the field's evolution and highlights the academic and policy study aimed to systematically map the intellectual structure and research trends in the field of behavioural economics and financial decision making from 2015 to 2025. The objective of this study was to provide quantitative overview of the fields evolution and identify future research direction. The publication gained momentum from 2023. The study found out Masaryk University, Czechia (35 articles), Roma Tor Vergata University, Italy (27 articles) and Zagreb University, Croatia (24 articles) as top 3 most prominent affiliated institution. Most relevant authors are Nicolini G, Kumar S and Zhang Y. These results shows that the research Most productions came from the United States, India and China out of which most cited countries remain United States and India. The analysis highlighted the International Journal of consumer studies and advances in neural information processing system as the core publishing output. The present topic is an emerging sector of research. Future work should broaden geographical participation and integrate behavioural insights with sustainability and inclusive policy frameworks. The strengthening links to financial literacy and long term good will ensure that behavioural finance research not only advances but also helps the overall benefit and development of the society

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